Outline of the topic and research project

The geometry of subvarieties on abelian varieties

This doctoral project studies subvarieties of complex abelian varieties through the conormal cycles associated with them. The main objective is to compute the dimensions and homology classes of new subvarieties produced by convolution and to explain their relation to monodromy and representation theory. Characteristic cycles provide the connection: they translate constructions involving perverse sheaves into conic Lagrangian cycles in the cotangent bundle, where the group law of the abelian variety permits a direct geometric description.

Let Z be a subvariety whose conormal Gauss map is generically finite. A general fibre determines finitely many points of Z. Taking their weighted sums and varying the covector produces subvarieties denoted by $Z^{(m)}$, with irreducible components governed by monodromy orbits. The dimension and fundamental class of each support can be recovered from the Segre classes of its conormal cycle, expressed in the Chern-Mather convention used in the preliminary work. This makes characteristic-class calculations a practical tool for studying these otherwise implicit geometric constructions.

The work completed so far gives explicit formulas when the monodromy is the full symmetric group and, for symmetric subvarieties, the full signed symmetric group. These formulas express the relevant classes in terms of Pontryagin products of the original classes, with coefficients obtained by combinatorial inversion on partitions. Applications to curves in their Jacobians yield explicit dimension formulas. Monodromy calculations for Abel-Prym curves provide further settings in which the method applies.

The next principal problem is the case of the even signed symmetric group, the Weyl group of type D. Here the signed sums of points in a general Gauss fibre split into two orbits according to the parity of the number of minus signs. These are related to the two half-spin representations and define two conic Lagrangian cycles. A recent construction using Schoen surfaces in abelian fourfolds shows that their homology classes need not agree: the example has conormal Gauss degree eight, monodromy W(D4), and distinct degree-four homology components of the two cycles. Consequently, a type D calculation must retain information that distinguishes the two orbits.

I will investigate how to compute the sum and the difference of these two classes separately. Degeneration to a union built from a genus-two Jacobian and a product of curves provides an explicit setting for studying specialization. Particular care will be given to the multiplicities of the summation maps and to contributions over loci where the Gauss map is not finite. The goal is to determine which additional geometric or Hodge-theoretic data are needed beyond the formulas already available in types A and C.

A second objective is to study the dimensions of the two projected supports. Their possible equality is a separate question, since unequal homology classes do not imply unequal dimensions. More generally, I will examine how changes in monodromy affect irreducibility and dimension, using the degree-three tangent Gauss counterexample to test the limits of large-monodromy assumptions. The intended outcome is an extension of the existing computational framework to selected type D cases, explicit formulas or obstructions for half-spin classes, and a clearer account of the geometric information carried by characteristic cycles.